

AI OF GOD - 5

Shape-Based Animal Classification from Segmentation Maps

A diverse ensemble of fine-tuned runs at several resolutions plus frozen multi-resolution heads, selected with out-of-fold evidence and not by leaderboard probing.

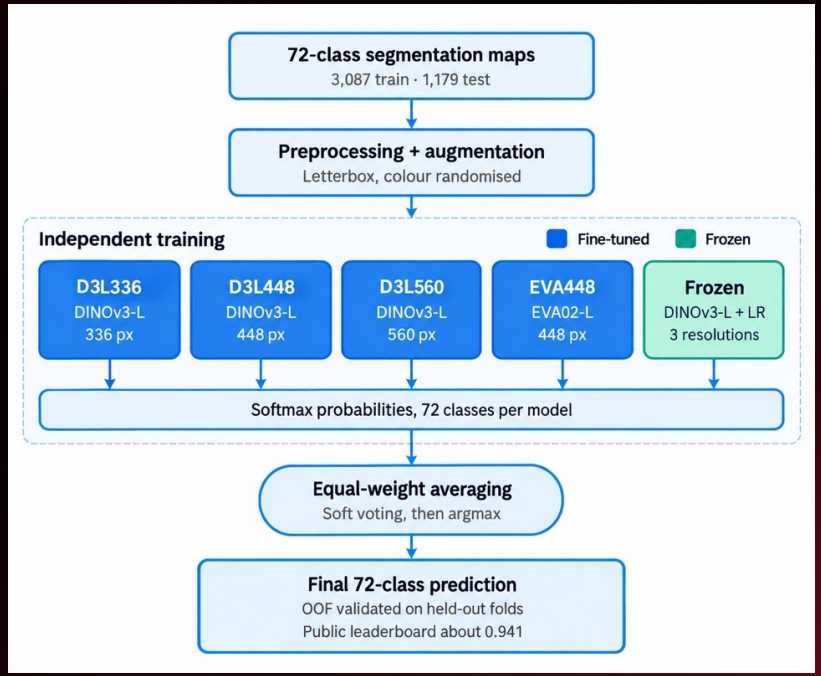
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Project Overview

This project builds a high-accuracy classifier for a colour-segmented animal dataset of 72 classes, using a deep ensemble of Vision Transformer based models.

Colours in the maps are arbitrary, so the models learn to rely on shape and layout rather than colour or texture.

Implemented in PyTorch with torchvision, PIL and scikit-learn; self-supervised DINO backbones are fine-tuned or frozen, and models are evaluated on macro F1 with out-of-fold validation.



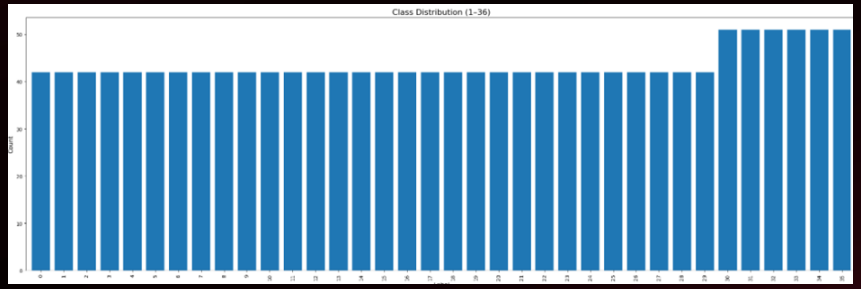
github.com/kavyanshusingh/ai-of-god

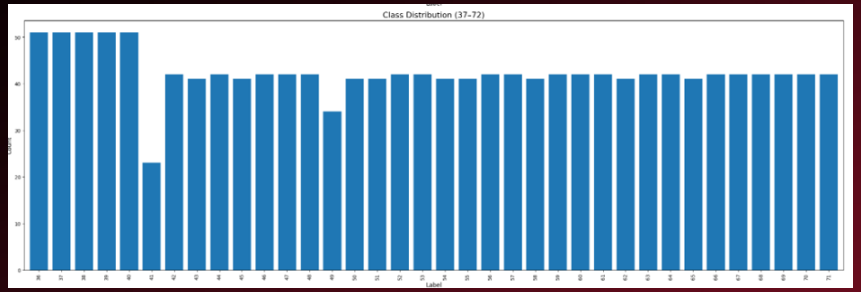
Problem and Dataset

Task: classify the animal in a semantic-segmentation map (animal + ground, sky, plants, objects).

- Colours are arbitrary, even within a class, so a colour-invariant model is needed
- Only ~43 images per class
- Strong pose and texture variation
- Metric: macro F1; public/private splits may cover different classes

Item	Value
No. of Train images	3,087
No. of Test images	1,179
No. of Classes	72
Mean / class	42.9
Max / class	51 (hedgehog)
Min / class	23 (jellyfish)
Median size	≈350×500 px

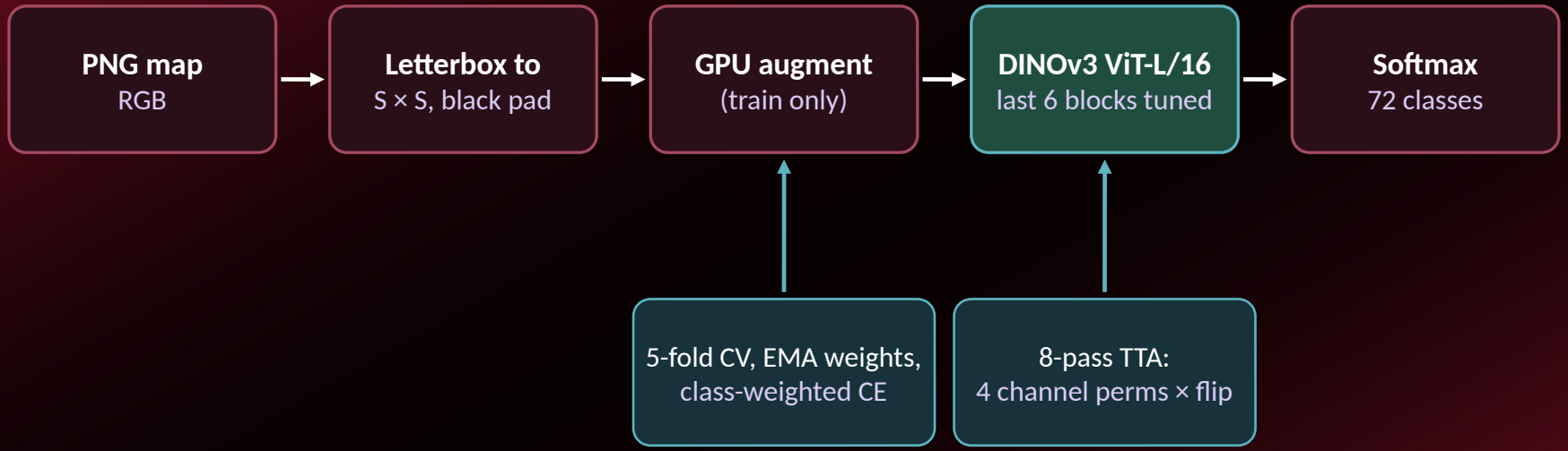




Roadmap of Experiments

Phase	Idea	Best result	Public LB
1	Silhouette / region maps + DINOv2 ViT-B	F1 0.604 (val)	–
2	Raw map + colour-destroying aug, ViT-B/14 @224	CV acc 0.949	0.810
3	DINOv2 ViT-L/14 @336, stronger aug + TTA	CV acc 0.979	0.839
4	CNN baselines: EfficientNet-B3, Stylized-IN ResNet-50	acc 0.94 / 0.84	–
5	**DINOv3 ViT-L/16 @336**	**CV F1 0.983**	**0.940**
7	Frozen DINOv3 multi-resolution heads + ensembles	41–46 OOF errors	+0.001
8	DINOv2 ViT-B/14 @224, colour jitter + flip aug	CV F1 0.946	0.856
9	DINOv3 ViT-B/16 @320, colour + geometric aug	–	0.889

Pipeline: From File to Prediction



Same 5-fold split used everywhere: StratifiedKFold(5, shuffle=True, random_state=0), so every run's out-of-fold (OOF) predictions are comparable and blendable.

Data Preprocessing Pipeline

1

Resize

Letterbox every map to the model's input size (S × S) with black padding, so the animal's aspect ratio is preserved.

2

Data augmentation

Applied on the GPU, per batch, during training only. Colour is destroyed by augmentation, not by preprocessing.

3

Normalize

ImageNet mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225]. Pre-trained models expect ImageNet-style inputs; this speeds up convergence and stabilizes training. The dataset was too small to use its own mean and std.

Data Augmentation (Applied on GPU, per Batch)

Data augmentation helps prevent overfitting by exposing models to more diverse training examples.

Group	Transform	Probability / range
Colour destruction	random 3×3 channel mixing + random monotone tone curve	60%
	random channel permutation	20%
	grayscale	20%
	inversion (1 − x)	15%
Degradation	5×5 box blur	30%
	Gaussian noise (std ≤ 0.05)	50%
Geometry	horizontal flip	50%
	rotation	±20°
	zoom / aspect jitter / translation	0.75–1.25 / ±10% / ±5%
	cutout (solid random colour)	30%, up to 20% side
Padding	letterbox kept black after colour changes	always

Not used: mixup, cutmix, random-resized crops, elastic warp, vertical flip.

Training Recipe and Test-Time Augmentation

Training

- Backbone frozen except last 6 blocks + final norm
- AdamW, weight decay 0.05
- Layer-wise LR decay 0.85 (top 3e-5), head 1e-3
- Class-weighted cross entropy, label smoothing 0.1
- Dropout 0.3, grad clip 1.0, fp16
- EMA of weights (evaluated copy)
- Batch 8; one fold ≈ 38 min @336

Inference

- 4 channel permutations × horizontal flip = 8 passes, softmax averaged
- Test = mean over the 5 fold models
- No multi-scale TTA

Why training loss looks high

With label smoothing 0.1 over 72 classes the loss floor is ≈0.74. Fold-4 training loss reached 0.80, so the model had fit the training set.

Phase 3: DINOv2 ViT-L/14 @336

Run	OOF	Public LB
ViT-B/14 @224 (Phase 2)	acc 0.949 / F1 0.948	0.810
ViT-L/14 @336, strong aug + 8-pass TTA	**acc 0.980 / F1 0.979**	**0.839**

+3.0
points in CV

+2.9
points on leaderboard

≈14
point gap still remained

Observation
CV and leaderboard moved together, but a gap of ≈14 points remained. We investigated it (next slide) before changing the architecture.

Phase 4: CNN and Stylized-ImageNet Baselines

Model	Setup	Acc	Macro F1
EfficientNet-B3 (ns, JFT)	fold 0, @336, 25 ep	0.940	0.937
EfficientNet-B3	fold 1	0.943	0.942
Stylized-ImageNet ResNet-50 (Geirhos et al.)	fold 1, @336, 25 ep	0.835	0.833
DINOv2-L (reference)	**fold 1**	**0.982**	**0.982**
DINOv2-L + SIN-ResNet50 blend	weight 0.1 / 0.3 / 0.5	0.984 / 0.982 / 0.976	0.984 / 0.982 / 0.976

- SIN-ResNet50 (strong shape bias) was 15 points weaker than DINOv2-L and did not help the blend (+1 image at weight 0.1, noise)
- No stylized ViT checkpoint was available; none was trained
- A plain (non-stylized) ResNet was not run separately

Phase 5: DINOv3 ViT-L/16, the Breakthrough

Backbone	OOF F1	OOF errors	Public LB
DINOv2 ViT-L/14 @336	0.979	–	0.839
DINOv3 ViT-L/16 @336	**0.983**	**52 / 3,087**	**0.940**

+≈10

leaderboard points from the backbone alone

What changed

Same augmentation, TTA and training recipe; only the backbone (and patch size 16, plus a rewritten GPU pipeline with a black-letterbox mask).

Likely cause: DINOv3's pretraining transfers better to unseen test maps. Not proven.

Resolution and Architecture Variants (Folds 0–1, 1,236 Rows)

Model	Errors	Hard-subset acc	Note
DINOv3-L @336	16	0.939	reference
DINOv3-L @448	**14**	**0.947**	**2 folds; fold ≈ 1.2–1.5 h**
DINOv3-L @336 + @448 (50/50)	**12**	**0.956**	**public score slightly lower**
EVA02-L @448	44	0.826	different family, much weaker
DINOv3-L @560	–	–	2 folds; no gain in any blend

Reading
Higher resolution helped OOF by 2 images but did not show on the leaderboard. EVA02 added diversity but was too weak to help blends.

Frozen DINOv3 Features at Many Resolutions

Res.	LR errors	kNN errors
336	71	98
448	63	99
560	59	94
672	56	90
784	**52**	**94**
all-res mean	54	90

Features: CLS + mean patch token, flip TTA; logistic regression (C=0.5) on standardised features. 5-fold OOF, 3,087 images.



Resolution keeps helping the frozen head; at 784 it equals the fine-tuned model. kNN was dropped.

Ensembles Tested (Out-of-Fold Errors, Lower Is Better)

All 3,087 rows

Ensemble	Errors	Fixed	Broken
D3L336 alone	52	–	–
D3L336 + frozen LR @448 (w=0.5)	48	–	–
D3L336 + frozen LR (336/448/672), w=0.5	46	–	–
D3L336 + D3L448 (reference)	45	0	0
+ frozen LR (560/672/784)	45	6	6
+ frozen LR (all res)	42	7	4
+ frozen LR @784	**41**	**10**	**6**

Folds 0–1 only (1,236 rows)

Ensemble	Errors
Reference	12
+ D3L560 + frozen	14
+ D3L560 + EVA448 + frozen	12
Five-way (submitted)	**12**

Differences of a few images are within noise: the ensemble has converged.

Final Submissions and Lessons

Selected finals

1. Five-way ensemble (equal weights): D3L336 (5 folds) + D3L448 (5 folds) + D3L560 (2 folds) + EVA448 (2 folds) + frozen DINOv3 LR (336/448/672)

2. D3L336 + frozen DINOv3 heads

The two differ by 5 labels.

Lessons

- Keep the raw map; destroy colour with augmentation, not preprocessing
- Backbone choice mattered most (DINOv2 → DINOv3: +10 LB points)
- OOF saturates near 0.98 and cannot rank near-equal models
- Macro F1 on a class-restricted split amplifies every error
- Ensembles of near-identical models converge; they add little

Thank You

AI OF GOD Team

We appreciate your time and attention. Our ensemble of fine-tuned and frozen DINOv3 models, selected with out-of-fold evidence, achieved robust predictions. We look forward to exploring future improvements and refined methodologies.

github.com/kavyanshusingh/ai-of-god